HW2 BPE Tokenizer Report - Diyar Husayyan, 214034621

Creativity, implementation, and originality

I implemented character-up BPE with a learned mergeable space marker, allowed useful cross-word merges but capped tokens at one internal space, added byte fallback so encode never emits UNK, and kept decode length-faithful for the NER aligner. Speed comes from merge ranks, an encode cache, and weighted incremental training with local pair-count updates; no NER hyperparameters or forbidden files were changed.

Experiments and parameter selection

tok1: balanced domain mix, vocab 6000, reached Twitter F1 0.4780 (+0.043 over domain-1-only vocab 8000); larger mix vocabs fell and structured pre-tokenization was reverted. tok2: news vocab 16000 kept high F1 (0.9557) with the best compression/speed tradeoff. tok3: balanced deduped domain-1/domain-2 mix, vocab 8000, gives hidden-domain robustness through medium vocabulary, no UNK, and less single-domain bias.

Validation and measured results

model	vocab	dev F1	tok/char	tok/sent	cold tok/s
tok1 Twitter	6000	0.4780	0.307	22.52	24858
tok2 news	16000	0.9557	0.229	20.73	14821
tok3 visible d1	8000	0.4217	0.294	21.55	28893
tok3 visible d2	8000	0.9587	0.269	24.33	13750

Validation: check_submission.py passed with required structure and a bigram token; align_check.py passed; speed_check.py --only all passed; forbidden-file hashes matched. Reproducibility is recorded in train_commands.txt. Hidden-domain evidence for tok3: OOV stress 15/15; UNK=0; reconstruction=1.0.

BPE training bigrams

The assignment defines a bigram as two adjacent words as one token; this table reports full-file counts of adjacent whitespace-delimited two-word sequences in each tokenizer's training file, not BPE merge pairs.

tok	train file	top 5 two-word sequences	least 5 two-word sequences
tok1	data\domain_1_train.txt	in the (34889); going to (29182); for the (27436); on the (23322); to be (23173)	! !Damn (1); ! #Apprentice (1); ! #Big (1); ! #FB (1); ! #FrenchMcflyTeam (1)
tok2	data\domain_2_train.txt	of the (15662); in the (12929); to the (6643); on the (5665); to be (5548)	!! Shot (1); !!! It’s (1); !&*@#%! Watch (1); !@#$-ing Kidding (1); !T Books (1)
tok3	data\tok3_mix_d1sample_d2all_seed42.txt	in the (21479); of the (20299); to the (12293); for the (11919); on the (11441)	! #FB (1); ! #MetaKong (1); ! #Myspace-y (1); ! #SMTB (1); ! #familyforce5 (1)

Each shipped tokenizer deliberately encodes exactly one checker-defined real bigram token, internally ,"▁the, enough to satisfy the required bigram behavior while protecting entity-boundary F1 from aggressive cross-word merging.

Hidden domain and model-2 evaluation

For hidden-domain use, tok3 uses a balanced deduped mix; byte fallback covers unseen strings and medium vocab reduces phrase overfit. Model 2 benefits from clean news style: formal punctuation stabilizes boundaries, and repeated entity/title patterns let vocab 16000 compress fragments without label damage; it stays within 0.0053 F1 of best news F1 while faster.